

ARIF KURNIA WIJAYANTO

M.Sc., Ph.D.

Lecturer & Researcher — GIS, Remote Sensing and UAV Applications for Natural Resources Management

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PROFILE

Geospatial scientist and lecturer specializing in GIS, remote sensing, and UAV-based monitoring for agriculture and natural resources management. Ph.D. in multi-sensor UAV platforms for agricultural damage assessment (Chiba University, Japan) with over a decade of experience leading research, teaching, and system-development projects funded by international agencies including JICA, UNDP, and the UK Space Agency. Licensed remote (UAV) pilot and certified GIS analyst.

EDUCATION

2020 – 2025

Doctor of Philosophy (Ph.D.) — Chiba University, Japan

Research topic: Application of a multi-sensor UAV platform for developing a damage assessment method in agricultural insurance.

2013 – 2016

Master of Science (M.Sc.), IT for Natural Resources Management — IPB University, Indonesia

Study interest: GIS-based Decision Support System and Expert System for Natural Resources Management.

2008 – 2012

Bachelor of Science, Agricultural Technology — Faculty of Agricultural Engineering and Technology, IPB University, Indonesia

Study interest: Informatics engineering for agriculture.

ACADEMIC & PROFESSIONAL EXPERIENCE

2020 – Present

Lecturer, Department of Forest Resources Conservation and Ecotourism, Faculty of Forestry and Environment, IPB University

Teach theory and practical courses in GIS and remote sensing to undergraduate and postgraduate students.

2016 – Present

Researcher (IT and GIS/RS Staff), Center for Environmental Research, Institute of Research and Community Development, IPB University

Manage IT infrastructure for online scientific journals; develop and manage the online system for the LAPAN-IPB Satellite (LAPAN-A3).

2018

Expert Team, Dynamics System Modelling — PT Geospasial Infodinamika (GeoID)

Project: Development of a spatial dynamics model for the Sorong Special Economic Zone (KEK Sorong), Geospatial Information Agency (BIG); modeled the effect of the zone on regional development.

2010 – 2012

Coordinator of Web and Multimedia Design, Engineering Design Club (EDC) — Department of Mechanical and Biosystem Engineering, Bogor Agricultural University

Managed the web design division's annual activity plan and coordinated the design crew.

2011

Internship Student — PT Hindoli (a Cargill company), Musi Banyuasin, South Sumatra

Studied implementation of the management information system for CPO production, from pre-harvest to post-harvest.

TEACHING EXPERIENCE

Course Name	Level
Environmental Spatial Analysis	Undergraduate
Mapping, GIS, and UAV for Environment	Undergraduate
Spatial Modeling for Conservation	Postgraduate
Spatial Modeling for Natural Resources Management	Postgraduate
IoT for Conservation	Postgraduate

RESEARCH INTERESTS

- GIS and remote sensing technology, especially for agriculture and natural resources management
- Web technology, especially for agriculture and natural resources management
- Decision Support Systems (DSS), Expert Systems (ES), and Artificial Intelligence (AI)
- Knowledge management systems for agriculture and precision agriculture
- Monitoring and control systems

SELECTED RESEARCH PROJECTS

Year	Project	Position	Funded By
2013 – 2014	Development of an expert system for broiler management in closed houses	Research Assistant	BOPTN
2016 – 2017	Rearrangement of the Environmental Quality Index (IKLH)	GIS & Remote Sensing Assistant	Ministry of Forestry and Environment
2016	Adaptation planning for climate change vulnerability, Kabupaten Indragiri Hilir, Riau Province	GIS & Remote Sensing Expert	Badan Lingkungan Hidup Daerah, Kab. Indragiri Hilir
2017 – 2020	Forest2020 — international research project	Researcher	UK Space Agency
2018	Spectral reflectance of LAPAN-IPB (LAPAN-A3) Satellite and Landsat 8 over agricultural area in Probolinggo, East Java	Principal Investigator / Team Leader	Osaka Gas Foundation of International Cultural Exchange (OGFICE)
2018 – 2020	Spatial-based fertilizer recommender for oil palm (collaboration with PT Pupuk Kaltim)	Web Developer	PT Pupuk Kaltim
2019 – 2020	National Monitoring System for Forest Land Cover and Strategic Commodities for Sustainable Landscape Development	Researcher, WebGIS & Mobile App Developer	UNDP
2019 – 2022	SATREPS: New Damage Assessment Process in Agricultural Insurance as Climate Change Adaptation for Food Security	Researcher, UAV Pilot	JICA

CERTIFICATIONS & LICENSES

Year	Certification	Issued By
2024	Remote Pilot (Refreshment)	Directorate General of Civil Aviation, Ministry of Transportation, Indonesia
2020	Remote Pilot	Directorate General of Civil Aviation, Ministry of Transportation, Indonesia
2018	GIS Level 6: Analyst (Decision No. 02.22.03/KEP/LSTPMPN/2018)	Lembaga Sertifikasi Profesi (LSP) MAPIN

TRAINING & PROFESSIONAL DEVELOPMENT

Year	Training	Hosted By
2024	Remote Pilot Course	TerraDrone Academy
2020	Basic Remote Pilot Course	Indonesia Aerosport Federation
2018	ISO 9001:2015	Cevral Consulting
2018	Hectare Indicators Training	Ecometrica, UK
2017	EO Lab Web Mapping Course	Ecometrica, UK
2016	ArcGIS Platform Technical Workshop	PT Esri Indonesia

LANGUAGES

Language	Speaking	Reading	Writing	Certification
Indonesian	Good	Good	Good	-
English	Good	Good	Good	ITP-TOEFL 570

TECHNICAL SKILLS

Office & Design: Microsoft Office (Word, Excel, PowerPoint, Visio, Access); CAD (AutoCAD, SolidWorks, CATIA)

Databases: SQL (PostgreSQL, MySQL)

Automation: n8n

GIS & Web-GIS: ArcGIS, QGIS, PostGIS, GeoServer, MapServer, Leaflet, OpenLayers

Programming: Javascript, R, Python

PUBLICATIONS

- Hartoyo, A. P. P., Karlinasari, L., Setiajiati, F., **Wijayanto, A. K.**, Rifana, H. Z., Madani, H. N., Satriawan, H., Rahmawati, R., & Siregar, U. J. (2025). Estimating Vegetation Density Dynamics, Tree Diversity, and Carbon Stock in the Agroforestry System of the Community Forest in Bogor Regency, Indonesia. *Jurnal Sylva Lestari*, 13(1). <https://doi.org/10.23960/jsl.v13i1.989>
- Rushayati, S. B., Prasetyo, L. B., Pramesti, T. V., Wijaya, F. H., & **Wijayanto, A. K.** (2025). Spatiotemporal analysis of land surface temperature and biomass changes in Nusantara Capital City: Challenges for forest city planning. *Ecological Engineering and Environmental Technology*, 26(10). <https://doi.org/10.12912/27197050/210101>
- Wijayanto, A. K.**, Prasetyo, L. B., Hudjimartsu, S. A., Sigit, G., & Hongo, C. (2025). Advanced BLB disease assessment in paddy fields using multispectral UAV data and patch fragmentation metrics. *Smart Agricultural Technology*, 10(August 2024), 100766. <https://doi.org/10.1016/j.atech.2024.100766>
- Setiawan, Y., Prasetyo, L. B., Pawitan, H., Liyanton, L., Syartinilia, S., **Wijayanto, A. K.**, Permatasari, P. A., Syafrudin, A. H., & Hakim, P. R. (2018). Pemanfaatan Fusi Data Satelit Lapan-A3/Ipb Dan Landsat 8 Untuk Monitoring Lahan Sawah. *Jurnal Pengelolaan Sumberdaya Alam Dan Lingkungan*, 8(1). <https://doi.org/10.29244/jpsl.8.1.67-76>
- Amalo, L. F., Permatasari, P. A., & **Wijayanto, A. K.** (2021). Carbon Monoxide Emissions Monitoring During Covid-19 Pandemic In Jakarta. 42nd Asian Conference on Remote Sensing, ACRS 2021.
- Nanda, M. A., **Wijayanto, A. K.**, Imantho, H., Nelwan, L. O., Budiastira, I. W., & Seminar, K. B. (2021). Multi-Criteria Decision Analysis Methods for Siting Energy Generation from Municipal Solid Waste: A Systematic Mini Review. 7th International Conference on Electrical, Electronics and Information Engineering, ICEEIE 2021. <https://doi.org/10.1109/ICEEIE52663.2021.9616640>
- Yusuf, S. M., **Wijayanto, A. K.**, & Sudrajat, N. F. (2022). Soil erosion prediction using GeoWEPP model in Cimanuk Hulu sub-watershed. *IOP Conference Series: Earth and Environmental Science*, 950(1). <https://doi.org/10.1088/1755-1315/950/1/012105>
- Nanda, M. A., **Wijayanto, A. K.**, Imantho, H., Nelwan, L. O., Budiastira, I. W., & Seminar, K. B. (2022). Factors Determining Suitable Landfill Sites for Energy Generation from Municipal Solid Waste: A Case Study of Jabodetabek Area, Indonesia. *Scientific World Journal*, 2022. <https://doi.org/10.1155/2022/9184786>

9. Prasetyo, L. B., Setiawan, Y., Condro, A. A., Kustiyo, K., Putra, E. I., Hayati, N., **Wijayanto, A. K.**, Ramadhi, A., & Murdiyarso, D. (2022). Assessing Sumatran Peat Vulnerability to Fire under Various Condition of ENSO Phases Using Machine Learning Approaches. *Forests*, 13(6). <https://doi.org/10.3390/f13060828>
10. Seminar, K. B., Nelwan, L. O., Budiastira, I. W., Sutawijaya, A., **Wijayanto, A. K.**, Imantho, H., Nanda, M. A., & Ahamed, T. (2023). Using Precision Agriculture (PA) Approach to Select Suitable Final Disposal Sites for Energy Generation. *Information (Switzerland)*, 14(1). <https://doi.org/10.3390/info14010008>
11. **Wijayanto, A. K.**, Junaedi, A., Sujaswara, A. A., Khamid, M. B. R., Prasetyo, L. B., Hongo, C., & Kuze, H. (2023). Machine Learning for Precise Rice Variety Classification in Tropical Environments Using UAV-Based Multispectral Sensing. *AgriEngineering*, 5(4). <https://doi.org/10.3390/agriengineering5040123>
12. Nanda, M. A., Sugandi, W., **Wijayanto, A. K.**, Imantho, H., Sutawijaya, A., Nelwan, L. O., Budiastira, I. W., & Seminar, K. B. (2023). The Waste-to-Energy (WtE) Technology to Support Alternative Fuels for Agriculture in the Context of Effective Solid Waste Management in the Jabodetabek Area, Indonesia. *Energies*, 16(24). <https://doi.org/10.3390/en16247980>
13. Suyamto, D., Condro, A. A., Prasetyo, L. B., Setiawan, Y., & **Wijayanto, A. K.** (2020). Assessing the Agreement between Deforestation Maps of Kalimantan from Various Sources. *IOP Conference Series: Earth and Environmental Science*, 556(1). <https://doi.org/10.1088/1755-1315/556/1/012011>
14. Permatasari, P. A., Amalo, L. F., & **Wijayanto, A. K.** (2019). Comparison of urban heat island effect in Jakarta and Surabaya, Indonesia. *Proceedings of SPIE*, 11372. <https://doi.org/10.1117/12.2541581>
15. Sujaswara, A. A., Setiawan, Y., Prasetyo, L. B., Hudjimartsu, S. A., & **Wijayanto, A. K.** (2019). Utilization of UAV technology for vegetation cover mapping using object based image analysis in restoration area of Gunung Halimun Salak National Park, Indonesia. *Proceedings of SPIE*, 11372. <https://doi.org/10.1117/12.2540566>
16. **Wijayanto, A.**, Hudjimartsu, S., Prasetyo, L., Setiawan, Y., & Suyamto, D. (2018). System using geonode as opensource platform. *Proceedings - 39th Asian Conference on Remote Sensing, ACRS 2018*.
17. **Wijayanto, A. K.**, Sani, O., Kartika, N. D., & Herdiyeni, Y. (2017). Classification Model for Forest Fire Hotspot Occurrences Prediction Using ANFIS Algorithm. *IOP Conference Series: Earth and Environmental Science*, 54(1). <https://doi.org/10.1088/1755-1315/54/1/012059>
18. Suyamto, D., Prasetyo, L., Setiawan, Y., & **Wijayanto, A.** (2017). Combining Projective Geometry Modelling and Spectral Thresholding for Automated Cloud Shadow Masking in Landsat 8 Imageries. *Proceedings - UKSim-AMSS 11th European Modelling Symposium, EMS 2017*. <https://doi.org/10.1109/EMS.2017.15>
19. **Wijayanto, A. K.**, Yusuf, S. M., & Pambudi, W. A. (2019). The Characteristic of spectral reflectance of LAPAN-IPB (LAPAN-A3) Satellite and Landsat 8 over agricultural area in Probolinggo, East Java. *IOP Conference Series: Earth and Environmental Science*, 284(1). <https://doi.org/10.1088/1755-1315/284/1/012004>
20. **Wijayanto, A. K.**, Seminar, K. B., & Afnan, R. (2016). Mobile-based expert system for selecting broiler farm location using PostGIS. *Telkomnika*, 14(1). <https://doi.org/10.12928/TELKOMNIKA.v14i1.2903>
21. **Wijayanto, A. K.**, Seminar, K. B., & Afnan, R. (2015). Suitability mapping for broiler closed house farm using analytical hierarchy process and weighted overlay with emphasize on environmental aspects. *International Journal of Poultry Science*, 14(10). <https://doi.org/10.3923/ijps.2015.577.583>